

Alfred Amafeku

Technical Associate | STEM Educator

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Bugolobi - Uganda

Alfred

SUMMARY

STEM Educator and Technical Associate with over 5 years of experience supporting Robotics, prototyping, and maker-based learning environments. Skilled in maintaining and troubleshooting 3D printers, CNC Machines, Embedded systems, Sensors and electronics equipment while guiding students through hands-on design and innovation projects. Experienced in digital fabrication tools, Hardware and software programming systems, CAD software, rapid prototyping workflows, and STEM workshop facilitation. Passionate about empowering young innovators through practical experimentation, creative technology and project-based learning.

Experience

Fundi Robotics Limited

Technical Associate - (Fundi@Work)

Jan 2023 – Present

Kampala, Uganda

- Implemented hands-on robotics and technical training programs for university interns and out-of-school youth to strengthen practical engineering and problem-solving skills.
- Maintained, troubleshoot, and repaired robotics equipment, 3D printers, CNC Machines and electronic tools to ensure uninterrupted and safe lab operations.
- Designed and developed robotics and coding learning modules focused on practical application and innovation
- Facilitated, supervised and mentored students in building functional robotics and electronics projects through guided prototyping and experimentation.
- Participated in a Design Thinking training facilitated by Design Without Borders focused on innovation and user-centered product development.
- participated in Capacity building sessions such as Internet of Things, PCB development, Advanced CAD and Simulations
- Designed learning modules for both teachers and learners to enhanced science education at primary and secondary levels of education

Fundi Robotics Limited

Teaching Associate - Robotics (Fundi@School)

Jan 2020 – Dec 2022

Kampala, Uganda

- Developed hands-on learning aids, such as models of the logic gates, digestive system, reproductive organs, electric bell to enhance science teaching at primary schools.
- Conducted Training of primary and secondary Teachers aimed towards tool integration in the classroom.
- Conducted interactive Robotics training and Computer programming sessions at schools.
- Designed and wrote learning modules for the Robotics, Coding and Enhanced Science Curricula.
- Integrated digital tools (Scratch, PictoBlox, Arduino, Tinkercad, Google Meet, Google classroom) to enhance computational thinking and problem-solving skills of students.
- Provided one-on-one support to students challenged by advanced science concepts.
- Participated in the monitoring and evaluation of the progress of children in learning their centers.

St Henry's College - Namugongo

Teacher of Physics and Mathematics - School Practice

June 2019 - Aug 2019

Namugongo - Uganda

- Served as the Class Teacher of Senior Six - Sciences
- Designed Schemes of Work out of the National Curriculum and innovative lesson plans to deliver Mathematics and Physics lessons
- Assessed and kept records of student progress through tests, assignments and practical exercises.
- Participated in professional development to stay up to date with advances in the education of Physics, Mathematics and STEM.
- Collaborated with other subject teachers to improve learning experience for students.

Education

WorldQuant University

Master of Science in Financial Engineering

2024 - 2025

Remote , <https://www.wqu.edu>

Mbarara University of Science and Technology

Bachelor of Science with Education - Physical

2016 - 2019

Mbarara , Uganda

Luzira C.O.U Senior Secondary School

Uganda Advanced Certificate of Education

2014 - 2015

Kampala , Uganda

Luzira C.O.U Senior Secondary School

Uganda Certificate of Education

2010 - 2013

Kampala , Uganda

Luzira C.O.U Primary School

Primary Leaving Examination

2009

Kampala , Uganda

Technical Skills

Prototyping 3D Printing, CNC Machines operation, Soldering, PCB Design, Welding, [Wowkwi](#))

Computer Programming Python,Django, JavaScript, PictoBlox, Arduino, HTML, CSS

Computer Aided Design SolidWorks, FreeCAD, Blender, Solid Edge, KiCAD, Altium, Proteus

Documentation $\LaTeX$, MS Office Suite, Google docs

Multimedia and Visual Design DaVinci Resolve, VSDC, Adobe Photoshop, Canva

Collaboration Google Classroom, Asana, Slack

Soft Skills

Traits Adaptability, Hardworking, Time management, Collaboration, Kaizen, Curiosity

Hobbies Soccer, Biking, Content Creation, Music, Movies, Traveling

Extra-Work Activities

Capacity Building and Professional Development

2024 - Present

- Learning Artificial Intelligence and Machine learning (Self-taught through various MOOCs).
- Actively upskilling in Unity 3D, Blender, and AI tools to expand capacity for creating immersive, interactive learning prototypes.

Amigos Football Club (Kampala - Uganda)

2024 - Present

- > As the captain of the club, I Organize and lead weekly training sessions and matches for a group of friends focused on physical fitness and team cohesion. Promote a healthy, active lifestyle while fostering teamwork, discipline, and sportsmanship.

STEM Club (Python and Robotics Training) (Kireka - Uganda)

2023 - Present

- > I design and facilitate 3-week intensive training programs in Python Programming and Robotics for secondary students during school holidays that take absolute beginners to building functional projects (GUI applications, robotic systems) in 15 days

STEM Voices Uganda (YouTube channel)

2025-Present

- > STEM Voices Uganda is a consortium of dedicated educators who believe in the power of science as a tool for sustainable development. This podcast highlights how science education drives innovation, fosters critical thinking, and contributes to the growth of Uganda and beyond. I edit the videos and participate in the content creation of the topics to be discussed.

Certificates

Strategies for Personal Growth and Development | Young African Leaders Initiative | May 2025

Asana Workflow Specialist | Asana | May 2025

Finalist | International Youth Mathematics Competition | November 2019

135 International course on Astronomy NASE-IAU | Network for Astronomy School Education | April 2018

Class President | Mbarara University B.Sc(Ed) -Physics Majors | June 2018

Sports Participant | Mbarara University of Science | June 2018

References

1. **Obote Geoffrey Tale**

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2. **Dr Ndugu Nelson**

Lecturer, Makerere University, Kampala, Uganda

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3. **Philemon Mindra**

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